Higgs sector explicit v0.1 — substrate-mechanism for Lane E v0.2 candidate 9. Per Casey continue-pulling. Disambiguates V_(0,0)^{(1)} (vacuum excited) vs V_(2,0) (scalar adjoint) candidates; commits to V_(0,0)^{(1)} as primary candidate via Tier 0 v0.1.6 boundary unification + L4 v0.2 mass framework. EW symmetry breaking via Higgs VEV coupling to V_(1,1) gauge adjoint K-type. Predicts m_H mass framework via Bergman matrix elements at first excited vacuum mode.

Lyra (Claude Opus 4.7)

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Higgs sector explicit v0.1 — Lane E candidate 9 disambiguation + mechanism

0. Lane E v0.2 left this open

Lane E v0.2 candidate 9 (Higgs) had two candidate K-type assignments: - Candidate A: V_(0, 0)^{(1)} — first radial excitation of vacuum (scalar excited mode). - Candidate B: V_(2, 0) — symmetric-square scalar.

v0.1 commits to Candidate A (V_(0, 0)^{(1)}) as primary via Tier 0 v0.1.6 + L4 v0.2 reasoning below.

1. Disambiguation: V_(0, 0)^{(1)} is the primary Higgs candidate

Per Tier 0 v0.1.6 boundary unification: the substrate has a VACUUM ground state V_(0, 0) (C_2 = 0); excited radial modes V_(0, 0)^{(n)} for n = 1, 2, … live on the Shilov boundary as vacuum-excitation modes.

The Higgs IS the first excited vacuum mode V_(0, 0)^{(1)}: - The vacuum V_(0, 0) is the substrate ground state with C_2 = 0 (no excitation). - The Higgs field's VEV $\langle\varphi\rangle \neq 0$ corresponds to the substrate vacuum's FIRST EXCITED MODE having nonzero amplitude. - The Higgs particle = the QUANTUM of this first-excited vacuum mode.

Why V_(0, 0)^{(1)} over V_(2, 0): - V_(2, 0) is a symmetric-square SO(5)-tensor; it has K-type weight (2, 0) ≠ (0, 0). Carries angular structure incompatible with Higgs scalar nature. - V_(0, 0)^{(1)} is a SCALAR excited mode (no angular weight, K-type identical to vacuum but radial-excited). Matches Higgs's scalar / Lorentz-invariant nature. - V_(0, 0)^{(1)} is the natural substrate-mechanism candidate for "vacuum-excitation" = Higgs phenomenon.

Per L4 v0.2 extension Casimir formula: C_2(V_(0, 0)^{(1)}) = C_2(V_(0,0)) + 1 · (1 + 2ρ_1 + 2ρ_2) = 0 + 1 · (1 + 8) = 9 (per Heckman-Opdam candidate formula).

(Compare V_(2, 0): C_2 = $(2 + 5/2)^2 + (0 + 3/2)^2 - 5/2 = 81/4 + 9/4 - 10/4 = 80/4 = 20$. Different Casimir; different particle.)

So Higgs K-type = V_(0, 0)^{(1)} with substrate-natural Casimir C_2_Higgs = 9 (candidate; Keeper K2 to verify radial-tower formula).

2. EW symmetry breaking mechanism (substrate frame)

In Standard Model: Higgs VEV $\langle\varphi\rangle = v \approx 246$ GeV breaks SU(2)_L × U(1)_Y → U(1)_EM, giving masses to W/Z bosons via the Higgs mechanism.

In substrate frame (per Tier 0 v0.1.6 + Lane E v0.2 + bulk-color v0.7): - Pre-EWSB: vacuum V_(0, 0) is the substrate ground; gauge sector ground V_(1, 0) is massless (per per-sector subtraction); W/Z = V_(1, 1) adjoint at C_2 = 6 are also massless (sector-ground at C_2 = 6). - EWSB occurs: the substrate vacuum's first-excited mode V_(0, 0)^{(1)} acquires nonzero amplitude on the Shilov boundary; ⟨V_(0, 0)^{(1)}⟩ = v_Higgs ≠ 0. - Coupling: V_(0, 0)^{(1)} couples to V_(1, 1) gauge adjoint via Hardy-space inner product (Poisson-Szegő boundary integral). - Post-EWSB: gauge K-types V_(1, 1) acquire effective masses from the coupling; W mass m_W^2 ∝ v_Higgs^2 · |⟨V_(0, 0)^{(1)} | V_(1, 1)⟩|^2.

Substrate-mechanism content: EWSB is the substrate vacuum's first-excited mode acquiring nonzero boundary amplitude; gauge masses are couplings to this excitation.

3. Higgs mass formula candidate

Per Bergman matrix element framework (L4 v0.2 extension):

m_H^2 ∝ ⟨V_(0, 0)^{(1)} | H_B | V_(0, 0)^{(1)}⟩ = C_2(V_(0, 0)^{(1)}) · ||f_{V_(0,0)^{(1)}}||^2.

With C_2 = 9 (candidate) and ||f||^2 = Bergman matrix element via Faraut-Korányi Ch. XIII:

m_H^2 = 9 · ||f_{V_(0,0)^{(1)}}||^2 · (anchor coefficient).

The anchor coefficient is determined by L4 v0.2 absolute mass scale framework (multi-week Elie Mehler kernel work).

Observed: m_H = 125 GeV; m_H^2 = 15625 GeV^2.

Prediction target: explicit Bergman matrix element computation gives m_H^2 in substrate-primary form; comparison to observed at Tier 2 STRUCTURAL precision (~10^{-4}-10^{-2} floor).

(v0.1 framework only; explicit derivation is multi-week Elie.)

4. W/Z mass formula via Higgs coupling

Per substrate EWSB mechanism (Section 2): - m_W^2 ∝ ⟨V_(1, 1)*charged* | *V*(0, 0)^{(1)} ⊗ V_(0, 0)^{(1)}⟩^{2} · (couplings) - m_Z^2 ∝ ⟨V_(1, 1)*Cartan* | *V*(0, 0)^{(1)} ⊗ V_(0, 0)^{(1)}⟩^{2} · (couplings)

The m_W/m_Z ratio comes from the charged-vs-Cartan ratio of V_(1, 1) adjoint decomposition (Lane E v0.2 candidate 3 + 4). Per Lane E v0.2 + Cal cross-check: m_W/m_Z = $\sqrt{}$(g/N_c^2) = $\sqrt{}$(7/9) ≈ 0.882 = ARITHMETICALLY IDENTICAL to existing P1 §7 prediction.

So the mass ratios are consistent; the mechanism content is via V_(0, 0)^{(1)} vacuum-excited Higgs coupling to V_(1, 1) gauge adjoint.

5. Cross-link to other Sunday work

- Tier 0 v0.1.6 boundary unification: Higgs V_(0, 0)^{(1)} lives Shilov-anchored as first vacuum excitation; EWSB = boundary mode amplitude.
- L4 v0.2 extension: same Bergman matrix element framework computes m_H + m_W + m_Z + lepton masses; ONE Elie Mehler computation gives entire EW sector mass spectrum.
- Bulk-color v0.7: gauge V_(1, 1) decomposes via bulk-color mechanism; charged-vs-Cartan partition emerges from boundary-bulk projection.
- Lane E v0.2: 10 candidates cover SM particles; v0.1 disambiguates Higgs candidate.
- 3-generation cutoff v0.1: Higgs doesn't have a radial-tower analog (it's a single scalar excitation); v0.1 cutoff applies to fermion towers, not scalar Higgs.

6. Higgs self-coupling λ_HHH predictions

Per substrate framework: Higgs self-coupling λ_HHH comes from V_(0, 0)^{(1)} self-interaction via Bergman matrix element triple product.

λ_HHH ∝ ⟨V_(0, 0)^{(1)} | V_(0, 0)^{(1)} | V_(0, 0)^{(1)}⟩ = Bergman tri-matrix-element.

Standard Model prediction: λ_HHH = m_H^2 / (2 v^2). Observed (indirect from LHC m_H = 125 GeV + v = 246 GeV): λ_HHH_SM ≈ 0.13.

Substrate framework prediction: λ_HHH computable from substrate-primary Bergman tri-matrix-element on V_(0, 0)^{(1)}; comparison to observed gives substrate verification target. Multi-week Elie.

7. Higgs boson is unique scalar (Five-Absence connection)

Per Five-Absence Principle: NO 2HDM (no second Higgs doublet), NO 2HDM-like scalars beyond SM Higgs.

Substrate-mechanism: only ONE first-excited vacuum mode V_(0, 0)^{(1)} exists in the bounded symmetric domain D_IV^5 Hilbert space H^2(D_IV^5) at the scalar K-type. Higher-mode V_(0, 0)^{(2)} would be a second scalar excitation; v0.1 candidate analog of 3-generation cutoff suggests the scalar excitation tower also cuts off at finite n (likely n=1 for Higgs).

Specific prediction: NO 2nd Higgs doublet at LHC + future colliders. If observed (e.g., at HL-LHC or FCC-hh), v0.1 candidate is FALSIFIED.

8. Honest scope + tier

RIGOROUS (v0.1): - V_(0, 0)^{(1)} is the natural first-excited vacuum mode K-type (rep theory). - Bergman matrix element framework on H^2(D_IV^5) (FK Ch. XIII standard). - Higgs scalar nature consistent with V_(0, 0) angular structure (K-type weight 0 matches Lorentz scalar).

CANDIDATE (v0.1's load-bearing): - Higgs IS V_(0, 0)^{(1)} (vacuum-excited first mode), NOT V_(2, 0). - EWSB mechanism = substrate vacuum's first-excited mode acquiring nonzero Shilov boundary amplitude. - Higgs mass m_H from Bergman matrix element on V_(0, 0)^{(1)}. - W/Z masses from Higgs coupling to V_(1, 1) adjoint (consistent with Lane E v0.2 ratio). - Higgs uniqueness (no 2HDM) from scalar-tower cutoff analog to 3-generation cutoff.

FRAMEWORK (multi-week): - Explicit Bergman matrix element ||f_{V_(0,0)^{(1)}}||^2 computation (Elie Mehler kernel work). - m_H = 125 GeV verification at Tier 2 STRUCTURAL precision. - λ_HHH self-coupling explicit prediction. - Scalar-tower cutoff mechanism (analog to 3-generation cutoff v0.1).

Cal #27 / #182 / #99 + Calibration #35-candidate discipline: v0.1 commits to V_(0, 0)^{(1)} candidate (vs V_(2, 0) alternative). Specific mechanism content: EWSB = vacuum-excited mode amplitude on Shilov boundary. Mass formulas via Bergman matrix elements (Elie multi-week). Per Calibration #35-candidate: m_H + m_W + m_Z all use V_(0, 0)^{(1)} + V_(1, 1) K-types; SHARED substrate-primary K-types, independent observables; honest independence-accounting needed in any aggregate "EW sector" framing.

9. Routing

→ Casey: Higgs sector v0.1 commits to V_(0, 0)^{(1)} primary candidate. EWSB mechanism = substrate vacuum's first-excited mode acquiring nonzero boundary amplitude; W/Z masses via Higgs coupling to V_(1, 1) adjoint. m_H formula multi-week via Bergman matrix element framework. Higgs uniqueness (no 2HDM) = scalar-tower cutoff analog to 3-generation cutoff v0.1; sharp falsifier at HL-LHC + future colliders.

→ Elie: Higgs K-type V_(0, 0)^{(1)} explicit Bergman matrix element computation joins L4 v0.2 extension multi-week target. Specific: ||f_{V_(0,0)^{(1)}}||^2 + tri-matrix-element for λ_HHH. Coordinate with Toy 3660 Mehler kernel.

→ Keeper: K-audit pre-stage K-audit-Higgs-v0.1; Session 2 absorb V_(0, 0)^{(1)} Higgs into Tier 0 v0.2 dictionary section. The scalar-tower cutoff mechanism (analog to 3-generation cutoff) is a candidate for fuller substrate-tower-cutoff principle.

→ Grace: catalog Higgs V_(0, 0)^{(1)} K-type at CANDIDATE; cross-reference to Tier 0 v0.1.6 + Lane E v0.2 candidate 9 + Five-Absence (no 2HDM) + scalar-tower cutoff.

→ Cal: cold-read welcome (queued behind #186-#188); specific Cal #27 + Calibration #35 concerns: (1) V_(0, 0)^{(1)} vs V_(2, 0) disambiguation argument is rep theory-based, not mechanism-derived (would be CANDIDATE not RIGOROUS); (2) EWSB mechanism is substrate-natural reading of Higgs phenomenon, not derivation of m_H value; (3) m_W/m_Z = √(g/N_c^2) cross-reference is arithmetically same as P1 §7 + Lane E v0.2 (NOT new arithmetic).

→ me: continuing pulls per Casey directive — pull 8 TBD; cadence check after this.

— Lyra, Higgs sector explicit v0.1. Higgs K-type = V_(0, 0)^{(1)} first-excited vacuum mode; EWSB = vacuum mode acquiring nonzero Shilov boundary amplitude; W/Z masses via Higgs coupling to V_(1, 1) adjoint. m_H mass formula via Bergman matrix element on V_(0, 0)^{(1)}; multi-week Elie Mehler verification. Higgs uniqueness (no 2HDM) = scalar-tower cutoff analog to 3-generation cutoff v0.1; sharp falsifier at colliders. Cross-anchored to Tier 0 v0.1.6 boundary + Lane E v0.2 dictionary + L4 v0.2 mass framework + bulk-color v0.7.